



UNIVERSITY OF
LEICESTER

School of Computing and Mathematical Sciences
CO7201 Individual Project

Interim Report

**HomifyOne: An AI-Powered Personalisation
and Workflow Management Platform for
New-Build Residential Developments**

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Chapter 1. Overview

My project on ‘HomifyOne: An AI-Powered Personalisation and Workflow Management Platform for NewBuild Residential Developments’ is progressing well. Overall progress across all the requirements sits at roughly 65%, with a few additional features added as the project progressed that were not part of the original requirements list. As of now, the full buyer journey is complete, and the main elements required for workflow management across users, covering developer review and approval, automated purchase order generation, and the supplier portal, are also completed to a good extent, with some of this work already ahead of the original schedule.

One of the core and novel features, the lifestyle questionnaire feeding a pre-trained sentence-transformer model and fetching ranked recommendation products through a FAISS similarity search, is completed, with some scope remaining to further optimise it. Several features originally scheduled for a later phase have also been brought forward. The overall project is in good shape. In this report, [Section 2](#) sets out the status of every individual requirement, and [Section 5](#) gives revised dates for everything still outstanding.

Chapter 2. Requirements

2.1. Essential Requirements

Multi-role secure authentication (JWT & bcrypt) *[Completed]*

- Implemented login, logout and OTP-based password reset for all roles.
- Passwords are hashed with bcrypt before storage.
- JWT carries the user's role and ID and is validated on every protected route via middleware.
- JWT is stored in localStorage and attached as an Authorization header on API calls, with an HttpOnly, SameSite=Strict cookie also set by the server as a fallback for requests that can't carry a custom header, such as direct PDF download links.

Challenges Faced: The real challenge here wasn't the login flow itself, but keeping buyer-specific data like basket contents, cached recommendations, questionnaire answers properly isolated between accounts, since these are cached in the browser's localStorage rather than tied to the session. On a shared browser, this could let one buyer's data carry over into the next buyer's login. This is handled in two places, AuthContext clears the relevant cached keys whenever the logged-in user ID changes, and BasketContext independently re-syncs its own state whenever it detects a different user, thus isolation doesn't depend on a single point of failure.

Role-based access control and dashboards *[Partial]*

- Buyer, Developer and Supplier dashboards are fully built.
- Each dashboard only shows the data relevant to that role.
- Access is enforced at the API level, not just hidden in the UI.
- Admin dashboard route exists and is protected, but its content is still a placeholder page.

Admin management of user accounts and product catalogue *[Partial]*

- Product catalogue management is complete. Now, admins can create, edit and delete products once their dashboard is ready.
- User account management hasn't been started yet.

Buyer personalised plot dashboard *[Completed]*

- Pulls the buyer's plot record and shows the property address.
- Shows current selection status.
- Live countdown to the deadline set by the developer.
- Buyers can view their property details and the amenities near their building location.

Room-by-room choices portal *[Completed]*

- Product choices included in the price of the home are grouped by room and sub-category, per plot type.
- Buyers can add, swap or remove a selection at any point up to submission.
- Buyers can view detailed product information to help them make more informed selections. Filtering options allow them to narrow down products based on relevant criteria, while pagination improves usability by organising large numbers of products into manageable pages.
- Rooms and spaces relevant to the buyer's chosen plot are only made visible to them.

Buyer questionnaire and AI recommendation engine *[Completed]*

- Questionnaire captures household composition, room usage and style preference.
- Answers are sent to a separate Python/FastAPI microservice.
- Buyer answers and the product catalogue are both converted into embeddings using all-MiniLM-L6-v2.
- A FAISS index returns the most semantically similar products, weighted by budget.

Challenges Faced: I had no prior experience with NLP or embedding-based search, so a good chunk of this milestone was research than coding, understanding how embeddings capture meaning rather than keywords, how FAISS searches efficiently as the catalogue grows, and how to turn a buyer's lifestyle answers into one query that maps sensibly onto product descriptions. Took a few rounds of adjustment to get the relevance right.

Dynamic live pricing engine *[Completed]*

- Totals recalculate instantly in the browser as priced products are added, removed or changed.
- Promo code discount and referral credit are applied before displaying the final total.
- Allowance progress bar accumulates across multiple orders instead of resetting each time.

Developer sets deadline, reviews, approves/rejects *[Completed]*

- Developers set and update a selection deadline per buyer.
- Full submission can be reviewed and approved/rejected.
- Approving triggers automated purchase order generation.

- Rejecting requires a written reason and resets the plot so the buyer can revise and resubmit.

Challenges Faced: Originally, submitting one order blocked a buyer from ever submitting again. The actual fix was narrower than first planned i.e., just block resubmission while a previous order is still awaiting review, and allow a new one once it's approved or rejected. The harder part turned out to be the pricing that follows from that, not the gating rule itself. The allowance progress bar has to add up expense across every past order without double counting, while the pre-submit screen still needs to total only the new items being submitted, so the two numbers had to be tracked and shown separately.

Automated purchase order generation grouped by supplier *[Completed]*

- On approval, items ordered by buyers are grouped by each product's assigned supplier.
- One purchase order is created per supplier, not per buyer.

Supplier restricted PO portal *[Completed]*

- Suppliers only see purchase orders assigned to their own account plotwise.
- Can acknowledge receipt, update fulfilment status, and set an ETA.
- Everything is scoped server-side to the logged-in supplier and not just filtered in the UI such that no tamper happens even if someone tries to.

Supplier invoice upload against purchase orders *[Completed]*

- Suppliers upload an invoice file directly against a specific purchase order.
- Stored in AWS S3 and linked back to that PO for developer review.

Developer PO and invoice management dashboard *[Completed]*

- One consolidated view of every purchase order and invoice across all buyers and suppliers.
- Developers can view or download the invoices.
- Developers can mark each invoice as paid or flagged.

In-app notifications for key workflow events *[In-progress]*

- Notifications work correctly for submission, approval, rejection, PO acknowledgement, invoice status, calendar and ETA updates.
- Scheduled reminders (cron jobs) for time-based events are in progress. For example, deadline reminders at 7/3/1 days out, overdue-selection flagging for developers, stale purchase order alerts for suppliers and developers

2.2. Desirable Requirements

Real-time in-app messaging via Socket.io *[Not Started]*

- Socket.io is already set up on both client and server.
- Currently only used to push notification events, yet to build chat data model, routes and UI.

Calendar synchronisation for meetings & deadlines *[Partial]*

- Replaced the planned Google Calendar API with an internal calendar model.
- Added an ICS export and subscription feed instead in the server side, so buyers and developers can add events to whatever calendar app they already use. UI for this implementation is in progress.

Supplier ETA synchronisation to developer's calendar *[Completed]*

- When a supplier sets or updates an ETA, it's added to the same internal calendar.
- Developer sees supplier ETAs alongside their own deadlines and meetings.

Developer interactive map *[Completed]*

- Built with Leaflet which shows every development a developer is assigned to, as a pin.
- Clicking a pin opens a panel listing plots, buyers and selection status.

Standard choices comparator *[Completed]*

- Buyers can select two or 3 included choices products in the same category and compare them side by side.

AI buyer assistant *[Partial]*

- AI microservice already has a working chat endpoint using FAISS search over the catalogue but it's yet connected to the Node server or a buyer-facing chat widget.
- The harder retrieval part is partially done what's left is wiring it up.

Challenges Faced: Earlier while drafting the preliminary report I set out two options to compare: retrieval-only, or a full RAG pipeline with Gemini API generating responses on top. Built and proved out the retrieval-only version first, since it can only ever return real catalogue items, never invented ones. Still deciding whether the generative layer is worth adding, or whether retrieval alone already covers most buyer questions well enough.

PDF selection summary on approval *[Completed]*

- Generated automatically on approval and uploaded to AWS S3 bucket which can be downloaded by both buyer and developer.
- Used PDFKit instead of the originally planned Puppeteer as it is lighter weight and no headless browser is needed.

2.3. Optional Requirements

AI invoice summariser *[Not Started]*

- Yet to start working on this.

AI warning system for missing/inconsistent choices *[Dropped]*

- Dropping this feature due to time constraints and its lower priority compared to the other core features.

Supplier invoice history and payment status dashboard *[Completed]*

- Suppliers can see every invoice they've submitted, along with its status of submitted, reviewed or paid.
- Separate menu for these in the supplier dashboard is yet to be created

Admin analytics dashboard *[Not Started]*

- Yet to start working on this.

Email notifications for all workflow events *[Partial]*

- Only OTP emails are implemented so far, via Nodemailer through Gmail.
- SendGrid and event-based emails (order submitted, approved, PO acknowledged, etc.) haven't been added yet.

The following requirements or enhancements were not provided in the preliminary report but added as the project progressed:

Top-up ordering model *[Completed]*

- This lets buyers submit additional selections after an existing order is approved or rejected, rather than being blocked entirely

Promo code and personalised credit system *[Completed]*

- Buyers get a fixed cash-off amount (between £50 and £250, based on 3% of their extras budget) automatically as part of personalisation credit when they finish the questionnaire.
- The credit is taken off first, then the 10% promo discount is worked out on what's left. This way, both rewards actually add up instead of one cancelling out the other.
- Both the credit and the promo code only work on a buyer's very first order, even if they place more orders later. The system checks this against their real order history, not just what the app remembers on their device.
- The server double-checks the credit amount against what the buyer is actually owed before applying it, so it can't be tricked into giving the discount twice.
- This was included in this project as part of a gamification technique which encourages buyers to complete the lifestyle preference questionnaire.

Buyer portal enhancements *[In-progress]*

- Included a Selection summary page that lists all past orders with downloadable selection summary PDFs
- On the buyer dashboard, added an order progress tracking bar.

Delivery status notification workflow *[In-progress]*

- Once every supplier invoice for a plot has been submitted, the developer can send a text progress update directly to the buyer via in-app notification.
- This feature will be enabled only when all the suppliers deliver the order items to the plot site and submit their invoices in their respective portals.

Chapter 3. Changes to initial plan

A few implementation details have diverged from the preliminary report's technical specification, all these are deliberate decisions and not accidental. Document storage now runs on AWS S3 while Cloudinary is used for storing just the product images. PDF generation uses PDFKit instead of Puppeteer and so this avoids running headless Chromium in production and is sufficient for the structured selection-summary documents required.

State management uses the Context API now. Redux Toolkit was dropped as unnecessary overhead for the application's current scope. The Google Calendar API integration described in the preliminary report has been replaced with an internal CalendarEvent model plus an optional ICS export and subscription. This approach was already identified in the project's risk plan as a fallback if the OAuth configuration required too much time. The fallback was introduced slightly earlier than expected, thus allowing development to remain focused on completing the more important buyer-facing features.

Priorities were also reshuffled. The full procurement pipeline under Milestone M4, and several Milestone M5 items including the map, comparator and PDF summary, were pulled forward and completed ahead of schedule, while admin-side work such as user management, the admin dashboard and analytics was pushed back, since it does not block any buyer, developer or supplier workflow.

Four features outside the original requirements list were added in response to needs surfaced during development.

Chapter 4. Key Achievements

- **AI recommendation pipeline:** Built the AI recommendation pipeline, which was honestly the biggest unknown going into the project. Earlier I flagged NLP and semantic search as an area I had no prior experience in, but I ended up using a real sentence-transformer model and FAISS pipeline that's now live and gives ranked, budget-aware recommendations based on the buyer's lifestyle questionnaire.
- **Procurement Pipeline:** Finished the entire procurement pipeline a week ahead of schedule. This included automated purchase order generation grouped by supplier, the supplier PO portal, invoice upload, and the developer's combined PO/invoice dashboard, which gave me extra breathing room for the remaining work.
- **Calendar Contingency:** Turned a risk into a feature before it even became a problem. Instead of relying on Google Calendar integration, I built a fully working internal calendar with .ics export/subscription support (whose UI implementation is pending) which is exactly the fallback I suggested earlier in the risk plan but done proactively rather than as a reaction.
- **Scope Extensions:** Went beyond the original scope where it made sense. As development progressed, real needs came up, so I added a top-up ordering model, a promo code and personalised credit system, further enhancements on buyer portal enhancements (in-progress) and delivery status notification workflow (in-progress).
- **Role-Based Security:** Tightened up role-based data isolation across all four stakeholder roles. Targeted testing uncovered some cross-account data leakage in the buyer basket and dashboard state, which I fixed and this in turn strengthened the security requirement that I had flagged earlier.
- **Dual-tracked pricing on resubmission:** One of the more difficult issues involved handling pricing when a buyer submitted additional selections. The allowance progress bar needed to show the buyer's total spending across all previous orders without counting the same items more than once. At the same time, the pre-submission page had to calculate only the cost of the new items being added in the current submission. Keeping these two totals separate and ensuring that one calculation did not affect the other required more work than the original gating rule itself.
- **Lightweight, production-lean architecture choices:** Several architectural decisions were made to keep the application lightweight and avoid unnecessary complexity. PDFKit was used instead of Puppeteer to generate selection-summary PDFs, as it does not require running a headless Chromium browser. The Context API was also chosen over Redux Toolkit for state management because the application's current size and complexity did not justify the additional setup and overhead of Redux. These choices helped keep the system simpler, easier to maintain, and more suitable for the project's present scope.

Chapter 5. Time-plan for the pending tasks

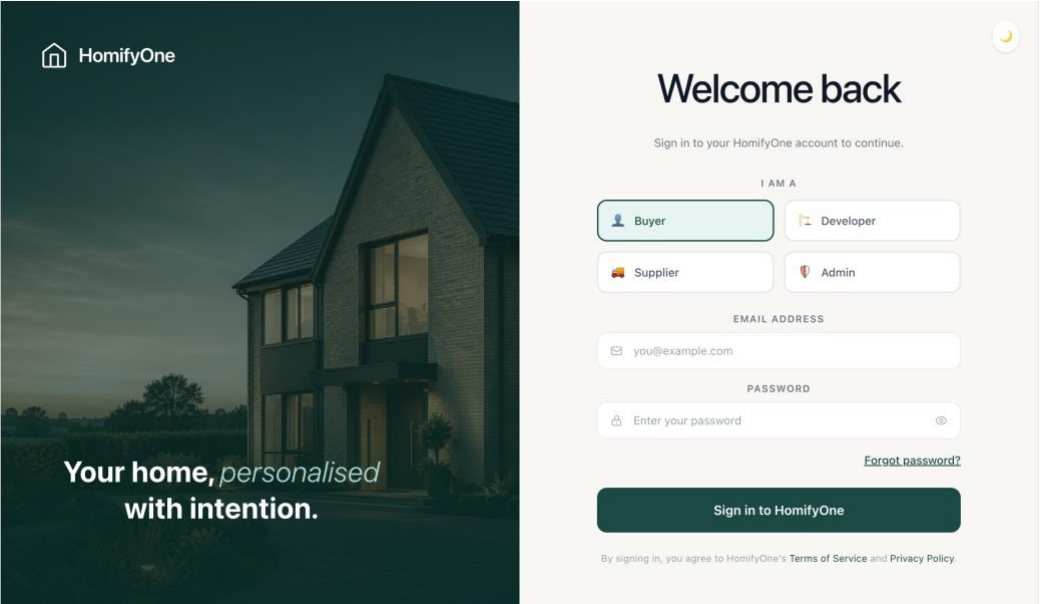
Here's the detailed time-plan for the tasks that are yet to be done:

S.No.	Task	Task Start Date	Task End Date
1.	Delivery status notification workflow	20/07/2026	22/07/2026
2.	Scheduled reminders using nodecron	21/07/2026	22/07/2026
3.	Buyer portal enhancements	23/07/2026	23/07/2026
4.	Real-time in-app messaging	22/07/2026	26/07/2026
5.	AI conversational assistant feature	27/07/2026	30/07/2026
6.	Admin governance panel & user management	31/07/2026	02/08/2026
7.	AI invoice summariser to extract key information	03/08/2026	05/08/2026
8.	Admin analytics dashboard	06/08/2026	07/08/2026
9.	Fallback Email Notifications for all key workflows	08/08/2026	08/08/2026
10.	Unit testing, Integration testing	09/08/2026	11/08/2026
11.	End-to-end system testing	11/08/2026	12/08/2026
12.	Final Report preparation	13/08/2026	30/08/2026
13.	Final report submission & Viva prep	30/08/2026	04/09/2026

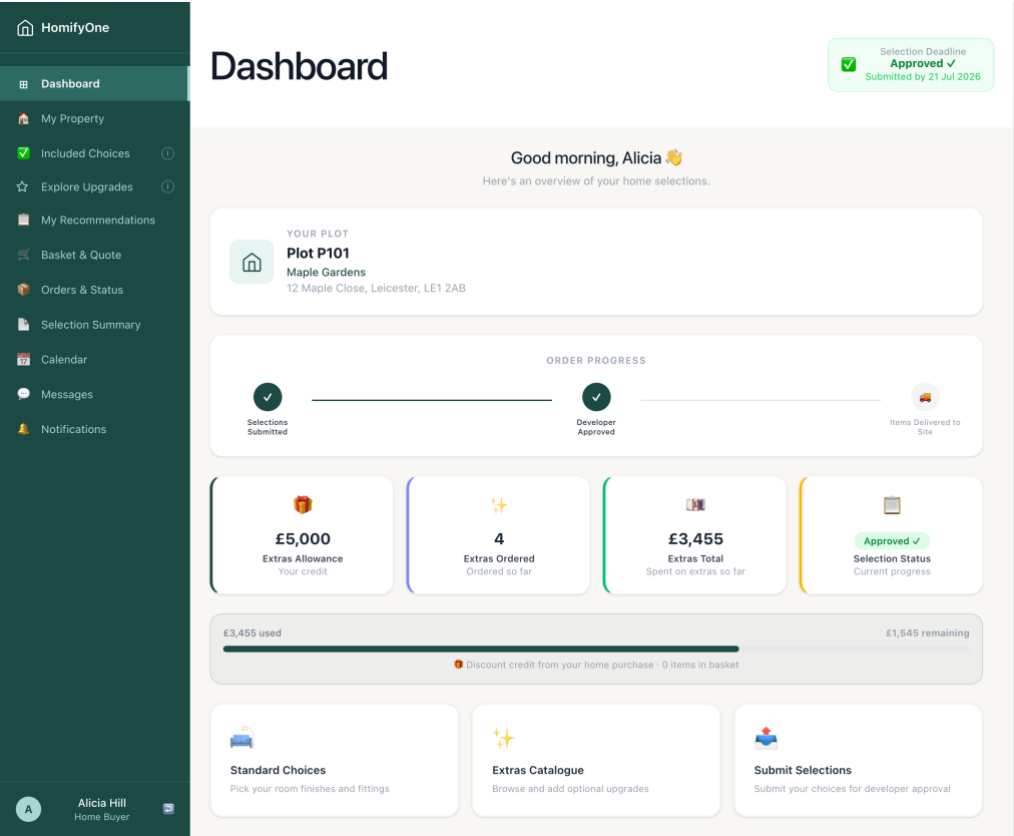
Chapter 6. Appendix

A. Buyer Journey

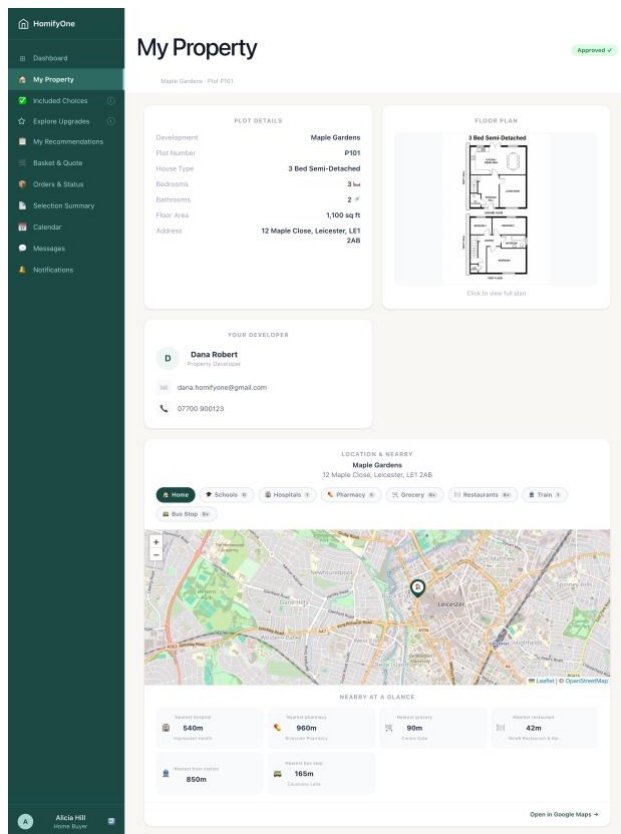
A.1. Login Page



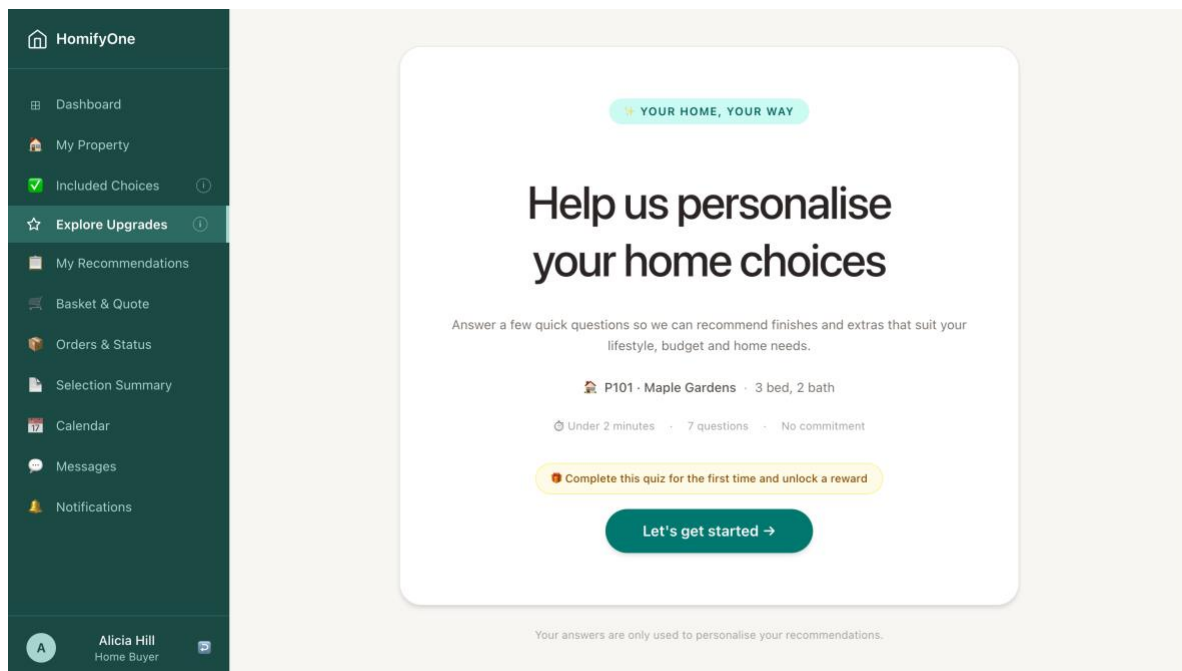
A.2. Buyer Dashboard



A.3. Buyer - My Property Page



A.4. Lifestyle Questionnaire Start Page



A.5. Buyer - My Recommendations Page

HomifyOne

Dashboard

My Property

Included Choices

Explore Upgrades

My Recommendations

Basket & Quote

Orders & Status

Selection Summary

Calendar

Messages

Notifications

My Recommendations

P101 - Maple Gardens

£1,540 of £5,000 remaining

Great! We're personalising your recommendations for family with young children, easy maintenance and long-term durability. We'll prioritise modern-style options and suggest upgrades within your budget of £0 to £10,000.

AllBedroomGardenKitchenLiving RoomSmart Home & Security

Sort: Best match£ Low → High£ High → Low24 results

45% match

Heritage Classic Cabinets

£2,175

Add to basket

42% match

Aura Integrated Wardrobe

£2,110

Add to basket

41% match

Scandi Chic Wardrobe

£2,080

Add to basket

40% match

Aurora Wide Spotlight

£165

Add to basket

40% match

Chic Mirror Wardrobe

£2,465

Add to basket

40% match

Milano White Sockets

£160

Add to basket

39% match

Luxe Panel Wardrobe

£2,620

Add to basket

39% match

Oslo Panel Radiator

£325

Add to basket

38% match

Aura Glass Wardrobe

£1,265

Add to basket

38% match

Pura Handleless Cabinets

£2,040

Add to basket

37% match

Linear Accent Lighting

£185

Add to basket

37% match

Horizon Integrated Fridge

£830

Add to basket

A.6. Buyer - Basket & Quote Page

HomifyOne

Dashboard

My Property

Included Choices

Explore Upgrades

My Recommendations

Basket & Quote

Orders & Status

Selection Summary

Calendar

Messages

Notifications

Basket & Quote

Plot P101
Maple Gardens

Review your selections before submitting

✓ Your previous selections were approved

You can add more upgrades/extras below and submit them as a new order.

Your upgrades/extras allowance

£5,000

Upgrades/extras total

£3,455

✓ £1,545 remaining from your allowance

Your basket is empty

Go to Included Choices or Explore Upgrades to add items.

Included ChoicesExplore Upgrades

A.7. Buyer - Order & Status Page

HomifyOne

Dashboard

My Property

Included Choices

Explore Upgrades

My Recommendations

Basket & Quote

Orders & Status

Selection Summary

Calendar

Messages

Notifications

Orders & Status

7 orders submitted

Submitted 16 Jul 2026

Approved

2 items

Within allowance

£315

Order total

View details →

Submitted 7 Jul 2026

Approved

7 items

Within allowance

£3,140

Order total

View details →

Submitted 7 Jul 2026

Changes Needed

6 items

Within allowance

£2,950

Order total

View details →

Submitted 7 Jul 2026

Changes Needed

7 items

Within allowance

£3,370

Order total

View details →

A.8. Buyer - Selection Summary Page

HomifyOne

Dashboard

My Property

Included Choices

Explore Upgrades

My Recommendations

Basket & Quote

Orders & Status

Selection Summary

Calendar

Messages

Notifications

Selection Summary

2 orders submitted

Order #2

Approved

Submitted 16 Jul 2026

2 items · £315

Download Summary

Order #1

Approved

Submitted 7 Jul 2026

7 items · £3,140

Download Summary

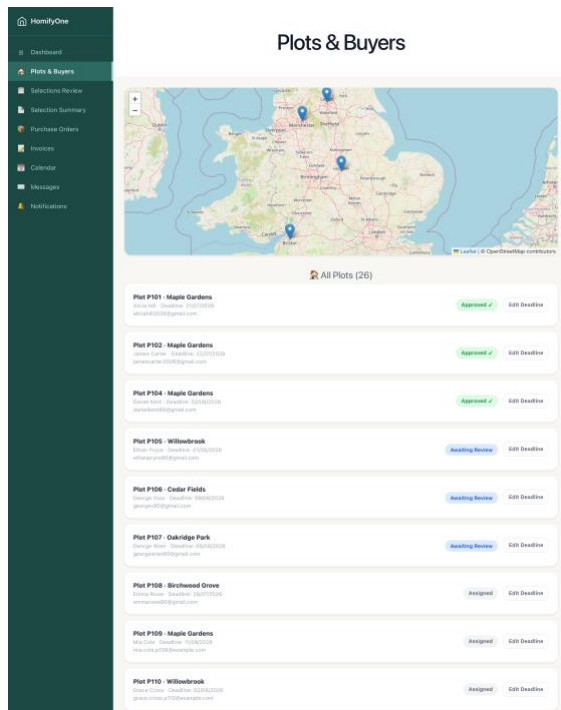
A

Alicia Hill

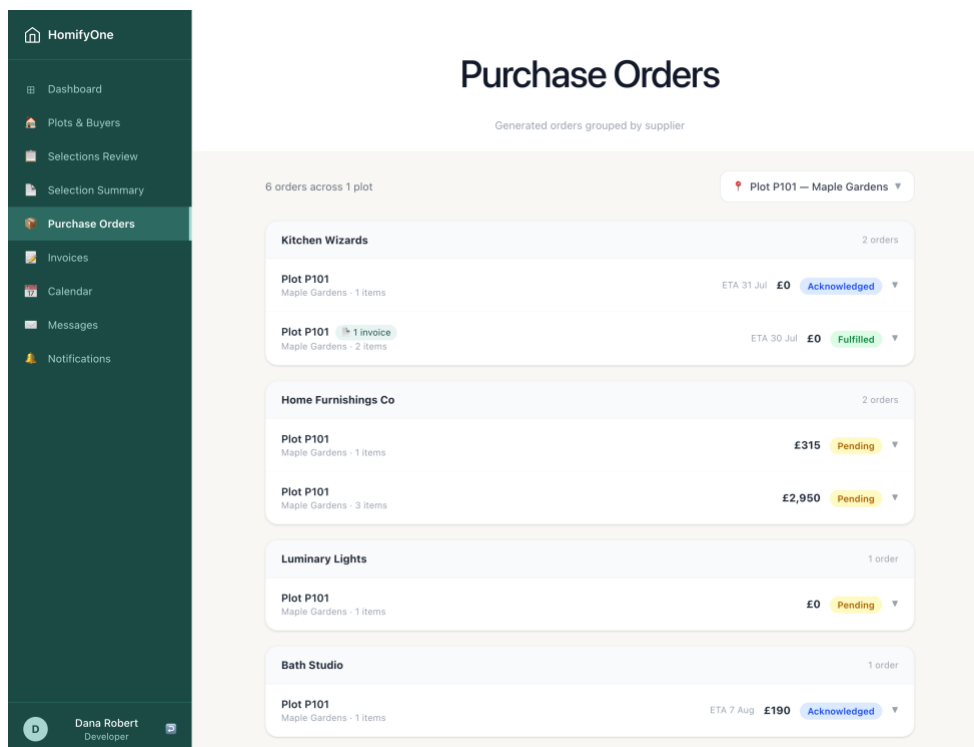
Home Buyer

B. Developer Journey

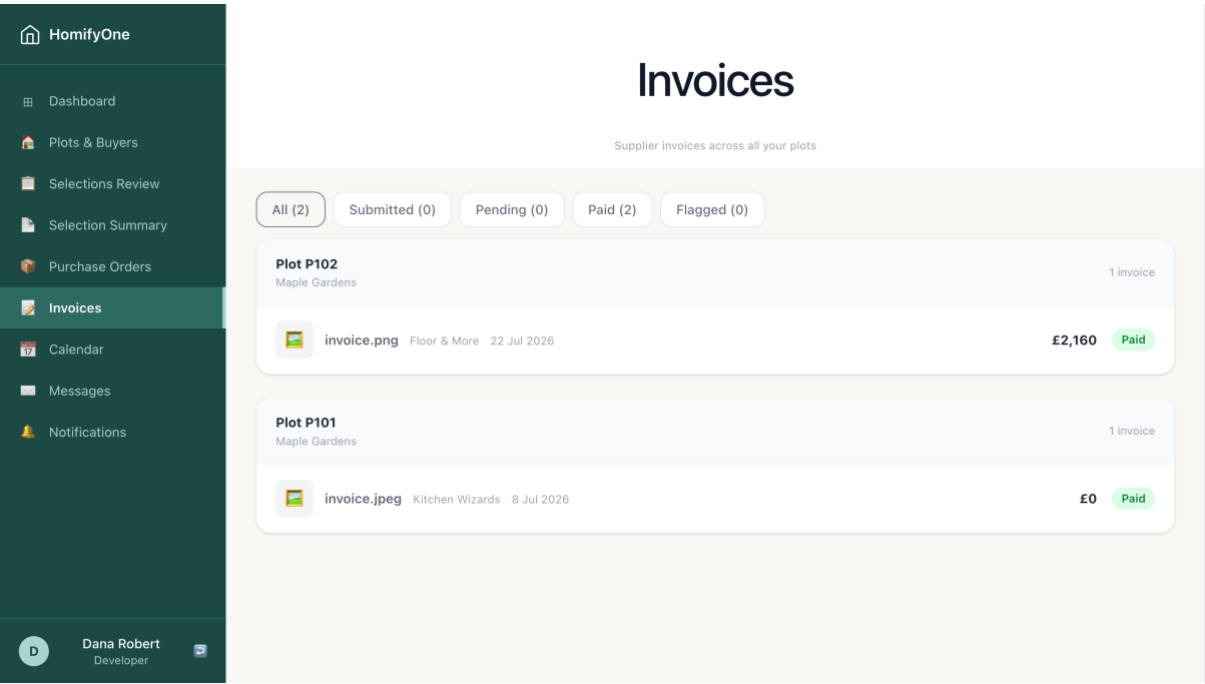
B.1. Developer - Plots & Buyers Page



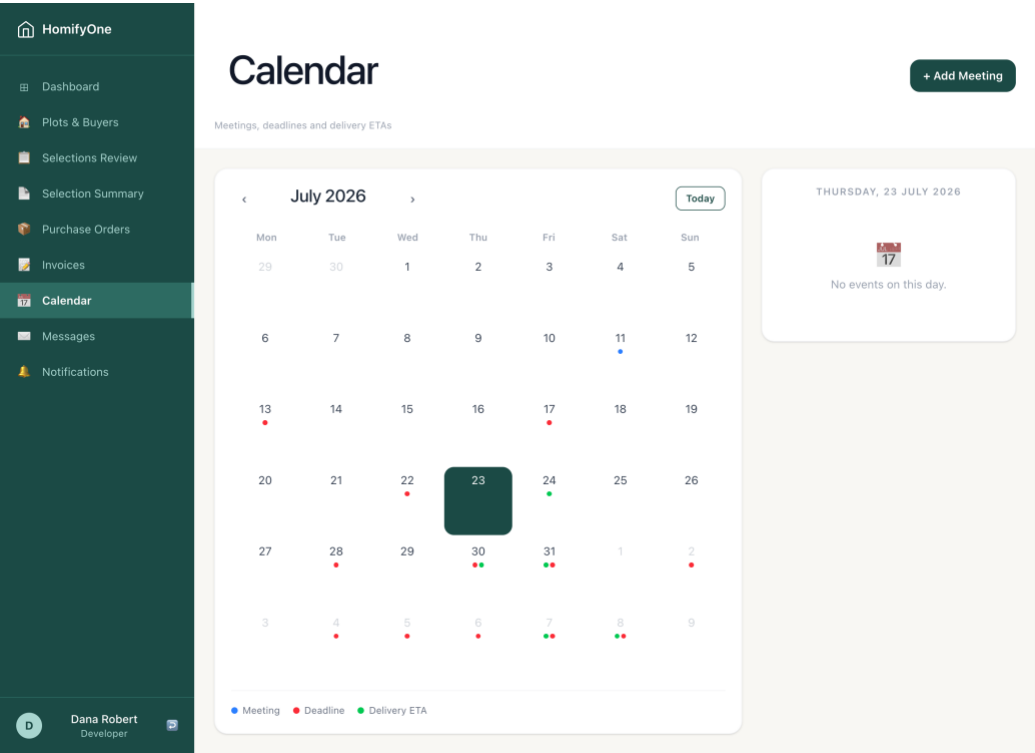
B.2. Developer - Purchase Orders Page



B.3. Invoices Page



B.4 Developer - Calendar Page



C. Supplier Journey

C.1. Purchase Orders Page

HomifyOne

Dashboard

Purchase Orders

Invoice History

Calendar

Messages

Notifications

Kitchen Wizards
Supplier

Purchase Orders

Orders assigned to you from developers

All (6) Pending (3) Sent (0) Fulfilled (1)

Plot	Developer	Items	Total	Status	Date	
P107 Oakridge Park	Dana Robert	1 items	£0	Pending	22 Jul 2026	View
P106 Cedar Fields	Dana Robert	1 items	£1,415	Pending	22 Jul 2026	View
P104 Maple Gardens	Dana Robert	1 items	£1,515	Pending	22 Jul 2026	View
P101 Maple Gardens	Dana Robert	1 items	£0	Acknowledged	16 Jul 2026	View
P102 Maple Gardens	Dana Robert	2 items	£0	Acknowledged	8 Jul 2026	View
P101 Maple Gardens	Dana Robert	2 items	£0	Fulfilled	7 Jul 2026	View

C.2. Invoice History Page

HomifyOne

Dashboard

Purchase Orders

Invoice History

Calendar

Messages

Notifications

Kitchen Wizards
Supplier

Invoice History

Your submitted invoices, grouped by plot

All (1) Submitted (0) Pending (0) Paid (1) Flagged (0)

Plot P101
Maple Gardens

1 invoice

invoice.jpeg

8 Jul 2026

£0

Paid

C.3. Notifications Page

